Finn Köhler

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Education

University of California, Berkeley — Berkeley, California, USA
Graduate Student - Computer Science & Engineering (Visiting Student) — *Aug. 2026 – Dec. 2026*
- Coursework (subject to availability): Vehicle Dynamics & Control (ME 236C), Advanced Robotics (CS 287), Algorithmic Human-Robot Interaction (CS 287H), Introduction to Embedded Systems (CS C249A), Dynamics and Control of Autonomous Flight (ME 236U).
- Focus Areas: Robotics, autonomous vehicles and real-time control systems.

University of Münster — Münster, Germany
Master of Science in Information Systems (Major in Data Science) — *Oct. 2025 – Sept. 2027*
- Current GPA: **1,5** (US equivalent ≈ **3.7**)
- Coursework: Unsupervised Learning, Optimization and Decision Making, Mining Massive Datasets, Computer Vision, Deep Learning, Supervised Learning
- Focus Areas: Machine Learning for autonomous systems.

Bielefeld University of Applied Sciences (HSBI) — Bielefeld, Germany
Bachelor of Engineering in Digital Technologies (Dual-Study Program) — *Aug. 2020 – Feb. 2024*
- GPA: **1,8** (US equivalent ≈ **3.5**); Thesis: **1,3** (US equivalent ≈ **3.8**)
- Thesis: Automated integration of TwinCAT Analytics Dashboard into existing industrial visualizations.
- Selected Coursework: Algorithms & Data Structures, Machine Learning, Cluster Computing, Big Data, Data Mining, Operations Research, Speech & Image Recognition, Quality Assurance for AI Systems.

Experience

Software Engineer (Working Student) — Oct. 2025 – Present
Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics) — *Münster, Germany*
- Building TwinCAT CoAgent MCP servers for the measurement suite, enabling automation across engineering tools and seamless communication between them.
- Engineering the TwinCAT Global Watchlist, a high-performance monitoring interface for real-time PLCs supporting **hundreds** of variables synchronized at PLC cycle time (down to **1 ms**).
- Implemented snapshot capture and restore, bidirectional write-back to live PLC runtimes, and flexible multi-format value rendering (hex, octal, binary, decimal).
- Developed a Visual Studio service enabling PLC self-registration and automated variable integration, letting end users add any PLC variable to the real-time watchlist via a single right-click.
- Contributing to the TwinCAT Scope Server for scalable real-time data analysis and diagnostics.

Software Engineer (Full-time) — Feb. 2024 – Oct. 2025
Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics) — *Verl, Germany*
- Built a high-performance, generic filtering framework in C# with a custom query language for tree views containing **hundreds of thousands** of items (full PLC variable lists with deeply nested arrays and structs).
- Leveraged reflection, Interop interfaces, and attribute-driven design to enable plug-and-play extensibility across modules.
- Optimized execution via aggressive caching and constraint-based evaluation, achieving real-time UI filtering with **millisecond latency**.
- Initiated the development of the TwinCAT Global Watchlist for real-time PLC diagnostics and visualization.

Dual-Study Software Engineer — Aug. 2020 – Feb. 2024
Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics) — *Verl, Germany*
- Bachelor Thesis (Grade 1,3 (US equivalent ≈ 3.8)): *Automated integration of the TwinCAT Analytics Dashboard into existing industrial visualizations.*
- Architected a synchronization framework unifying legacy visualization concepts with modern TwinCAT Analytics dashboards.
- Designed and implemented automated User Management configuration generation for secure server-side data access in TwinCAT HMI.
- Developed Windows kernel-mode drivers in C++ and industrial IPC diagnostics pipelines (CPU temperature, disk usage, system health) via the Beckhoff MDP library, asynchronous TwinCAT ADS communication, and direct memory reads.

Selected Projects

Part-Based Rigid Pedestrian Decomposition for Reconstruction of Autonomous Driving Scenes
PyTorch, 3D Gaussian Splatting, SMPL, Waymo Open Dataset 2026

- Developed **Method M**, a novel pedestrian model for the DriveStudio/OmniRe scene-graph framework, decomposing each pedestrian into **10** SMPL-driven rigid body segments with trainable per-segment pose residuals.
- Implemented SMPL forward kinematics, per-segment-aware Gaussian densification, and integrated the new node type across the data loader, trainer registry, and rendering pipeline (PyTorch, CUDA rasterizer).
- Improved Human-only PSNR by **+5.4 dB** over the Street Gaussians rigid-pedestrian baseline (**27.23** → **32.61**) on Waymo scene 23, with visibly coherent leg articulation in the reconstructed views.

Dartz - Multiplayer Darts Scoring Platform | *Next.js, ASP.NET Core, PostgreSQL, SignalR* 2024 – Present

- Architected a multiplayer scorekeeping platform with a dedicated ASP.NET Core game server using SignalR and WebSockets for real-time state synchronization across remote clients.
- Implemented authoritative game logic for "501" rules on the server-side with live matches supporting remote and local players, 3D dart models, and player performance analytics.
- **Live demo**

Handwritten Math Symbol Recognition + Grad-CAM | *Python, TensorFlow, CNN* 2023 – 2024

- Trained and evaluated a CNN to classify handwritten mathematical symbols (**80+** classes) using a Kaggle dataset with **100k+** images.
- Achieved **93.6%** test accuracy and analyzed failure modes via confusion matrices and misclassification clustering.
- Implemented Grad-CAM visualizations to explain model decisions and highlight salient regions used for classification.

Robotics club & Automated Pathfinding 2018 – 2020

- Engineered an autonomous robot capable of real-time pathfinding and object targeting using remote control and onboard logic.
- Organized STEM hands-ons for interested students in local schools.

Technical Skills & Leadership

Programming: C#/.NET (Advanced), Python (PyTorch, scikit-learn, pandas. NumPy, matplotlib), C/C++, TypeScript (React), SQL
AI & Data: Machine Learning, LLM pipelines, Deep Learning, Data Mining, Hadoop, Spark, MCP Servers, Agents
Automation & Real-Time Systems: TwinCAT PLC (IEC 61131-3), TwinCAT Analytics, TwinCAT HMI, TwinCAT ADS, Industrial IoT pipelines, Windows kernel-mode development, MQTT
Technologies: Git, Docker, Firebase, Hadoop, Spark
Methodologies: Agile/Scrum
Languages: German (Native), English (TOEFL iBT: **117/120**), Spanish (Basic)
Leadership: Handball Team Captain; Elected Member of Computer Science Student Council
Honors & Affiliations:

- **Fulbright Travel Scholarship Nominee:** Recognized for academic excellence and cross-cultural exchange (Declined to attend UC Berkeley).
- **German Informatics Society (GI):** Received the Award for Outstanding Achievement in Computer Science alongside an honorary membership.